

Day 01 – Assignment

Instructions:

- Execute the following tasks in Android Studio and paste screenshots of the code and the corresponding output in a document

Android Application Development – Fundamentals

1. Task 1

Change the application name to “TestApp”.

Hint: Check strings.xml and AndroidManifest.xml.

2. Task 2

Add a logo/icon to the application.

Hint: Use Android Studio → New → Image Asset to add the app icon.

3. Task 3

Create a simple Android app that displays “Welcome to Android Development” in the center of the screen.

Hint: Use a LinearLayout (android:gravity=“center”) or ConstraintLayout to position the TextView at the center of the screen.

4. Task 4

Set the background color of the screen to light blue (or some color).

Hint: Use the android:background attribute in the XML layout.

5. Task 5

Create a Greeting application where the user enters their name and clicks a button to display Hello, <name>!.

Hint: Use an EditText to enter the name, a Button to trigger the action, and a TextView to display the greeting. In Java, use `setOnClickListener()` to read the name and update the TextView.

6. Task 6

Modify Task 5 – Change the button color to dark green and its text color to black. Change the TextView text color to red.

Hint: Use `android:backgroundTint` and `android:textColor` for the Button. For the TextView, use `android:textColor`.

7. Task 7

Create a Random Dice Roller application that generates a random number between 1 and 6 whenever a button is clicked.

Hint: Use Random class to generate the random number and update the TextView inside `setOnClickListener()`.

8. Task 8

Create a Simple Calculator that performs addition, subtraction, multiplication, and division on two numbers.

Hint: Convert the input strings using `toIntOrNull()` or `toDoubleOrNull()` and perform the operation based on the selected button.

9. Task 9

Create an application with a TextView and a button. When the button is clicked, show or hide the TextView.

Hint: Use `setVisibility()`

10. Task 10

Create an Android application with three RadioButtons to select a favorite programming language. When the Submit button is clicked, display the selected language.

Hint: Use a `RadioGroup` to group the `RadioButtons` and `getCheckedRadioButtonId()` to find the selected option.